Available online at www.sciencedirect.com
SCIENCE @ DIRECT®

Electoral Studies 23 (2004) 279–290

Electoral Studies

www.elsevier.com/locate/electstud

General election forecasts in the United Kingdom: a political economy model

M.S. Lewis-Beck [a,*], R. Nadeau [b], E. Bélanger [b]

[a] Department of Political Science, University of Iowa, Iowa City 52242, USA
[b] Department of Political Science, University of Montréal, Canada

Abstract

With the notable exception of Mughan (1987), forecasting attempts in the United Kingdom have been solely concerned with developing popularity functions. This paper formulates a vote function model to forecast general elections in the UK. The model relies on three independent variables: approval of the government's record, inflation, and a variable controlling for the "cost of ruling." The model is estimated over the period 1955–1997 using a substantial six-month lead time. It performs reasonably well, and offers a plausible and parsimonious explanation of government vote support. The vote function is used to generate a forecast of the next British general election. It clearly predicts a majority government for Labour, assuming the election would be held in Spring 2001.

Keywords: Forecasting models; Vote function; Political economy; British elections

1. Introduction

The science of election forecasting has come of age. Twenty-five years ago, scientific models predicting elections were scarce, regardless of country. Now they proliferate. For virtually every established democracy, some political scientist or economist has put forward at least one such model. Work has been most extensive in the United States, the United Kingdom, and France. The models all posit an equation predicting government support, and estimate either a vote function or a popularity function.[1] The United States forecasting models focus exclusively on the vote func-

* Corresponding author. Tel.: +1-319-335-2358; fax: +1-319-335-3400.
E-mail address: michael-lewis-beck@uiowa.edu (M.S. Lewis-Beck).
[1] What Paldam (1981) originally referred to as the "VP functions."

0261-3794/$ - see front matter
doi:10.1016/S0261-3794(02)00071-9

tion, with the dependent variable commonly expressed as a party vote share.[2] The French forecasting models began as popularity functions, but there the move is to vote functions.[3] The British case, which concerns us here, has been considered almost entirely in popularity function terms, where the dependent variable is percent party support registered in a public opinion poll.[4] Below, we formulate a vote function model to forecast general elections in the UK. That model, while benefiting much from the popularity research, aims to avoid some of its forecasting difficulties. The equation, estimated on a time series of elections from 1955–1997, appears rather promising. As a challenge, we apply it to prediction of the upcoming parliamentary contest.

2. Popularity functions

While the dependent variable—popularity—is conceptually shared by electoral modelers in many democracies, its measurement varies from country to country. In the United States, for example, the standard Gallup question is "Do you approve or disapprove of how President Bush is handling his job?" In the United Kingdom, by way of contrast, the standard is "If there were a general election tomorrow, which party would you vote for?" Reflection on the two different wordings suggests why "popularity" is a more "popular" dependent variable in the British case. It directly assesses vote intention, whereas the US item is not that at all. Whiteley (1979) offered the first attempt to forecast UK government popularity, so measured. Utilizing a monthly time series of poll data ($N = 348$, 1947–1975) and Box-Jenkins techniques, he modeled current popularity as a dynamic function of past popularity, in a first-order autoregressive moving-average process. He attempted to predict the next actual vote intention score from the last pre-campaign poll (1951–1974). And, for 1979, he forecast a Labour victory (although in fact the Tories won).

Subsequently, much forecasting work has been done by Sanders (1991, 1993, 1995, 1996) who, in addition to modeling carefully the dynamic processes inherent in the popularity series, has tried to incorporate substantive independent variables, especially economic ones. In his thoughtful 1995 paper, he makes some important observations on forecasting with popularity functions: 1) the toughest test is forecasting the unknown future; 2) the preferred estimation strategy is the general-to-specific approach advocated by Hendry; 3) the preferred estimation period is over a single parliamentary term; 4) unpredictable events may overturn a forecast; 5) the "subjective economy" drives voter assessments (Sanders, 1995). Clarke and Stewart (1994), in something of a critique of Sanders, offer a different popularity function for forecasting, based on vector-autoregression (VAR) procedures and different inde-

[2] For recent examples, see the excellent collection of papers in Campbell and Garand (2000).

[3] See the current review in Lewis-Beck (2001).

[4] The noteworthy exception is Mughan (1987).

pendent variables, including Conservative support, national economic expectations and a misery index combining the unemployment and inflation rates.

Obviously, general election forecasting via popularity functions is a vibrant area of research in the UK. We introduce it as context, but do not propose to add to it, or solve its controversies other than indirectly. Our tack is different, with a different dependent variable, which we discuss below.

3. Vote functions

Mughan (1987) has written the lone paper on vote functions as forecasting tools in British elections. His main dependent variable (Vt) is actual party vote share won (in percent), over parliamentary contests from 1951–1983. He makes clear that his primary purpose is not explanation but forecasting—“predicting the outcome of an event before it occurs” (Mughan, 1987: 196). Therefore, independent variables were “sifted in a series of trial regression runs to determine which combination of them gave the best fit to the data” (Mughan, 1987: 198). Three forecasting models are estimated: 1) *incremental* (i.e., the vote is a function of its lagged value); 2) *opinion poll* (i.e., the vote is a function of the most recent popularity poll); 3) *economic* (i.e., the vote is a function of GDP in constant pounds and the unemployment rate in the year before the general election) (Mughan, 1987: 199). In comparing the models, he observes that “immediately striking is the relatively poor performance of the incremental model” (Mughan, 1987: 200). With respect to the opinion poll model, Mughan (1987: 200) says its weakness is that it “falls down on the criterion of allowing for a prediction *in advance* [sic] of the election [since] it depends on Gallup returns that become available only on the day of the election”. The economic model fares better in terms of goodness-of-fit (R - squared = 0.85) than the other two; however, when actually applied to generate an out-of-sample forecast of the 1987 result, it underestimates the government vote share by 3.3 percentage points, a miss equal to the polling model forecast error and bigger than the incremental model forecast error (Mughan, 1987: 201-202).

Mughan’s (1987) effort is exploratory, but path-breaking. He demonstrates that the idea of a forecasting vote function for Britain has a future, partly gleaned in his conclusion that “the simple economic model presented herein must be taken very seriously as a forecasting instrument for British general elections” (Mughan, 1987: 206). We turn to the development of this economic notion, along with a more general attention to model specification.

4. Economics and the British voter

A substantial literature shows that the British voter, like his or her counterpart in other Western democracies, takes economic conditions into account in casting the vote. Currently, the debate is not over whether economic voting exists. Rather, it is over what economic variables are relevant, and how and when they are perceived.

This controversy cannot be solved here, if for no other reason than our analysis is of aggregates, not individuals.[5] A recent summary of the work concludes that “voters reward government with their support if their economic prospects look good and if they perceive that unemployment and inflation are falling; they inflict punishment by withdrawing their support if expectations are falling or if they perceive that unemployment or inflation are rising…. [Furthermore], [u]nemployment perceptions track ‘real’ unemployment and inflation perceptions track ‘real’ inflation very well” (Sanders, 2000: 287, 290). For our purposes, it is important to note the weight given to inflation, for it emerges as our key economic independent variable.

The assembled data-set is a time series of 12 consecutive parliamentary elections, up to and including 1997. With such a small *N*, all possible economic variables cannot be included in a model, even experimentally, because of the very limited degrees of freedom. However, we do have complete observation on five macroeconomic variables that have been much studied in the British case: the unemployment rate, the inflation rate, the interest rate, growth, and a misery index (see Appendix for details). All correlate in the expected direction with the dependent variable of government party vote share (see Table 1).

Which seem the most promising candidates for testing the economic voter hypothesis? On the basis of the above theoretical assessment by Sanders, featured variables would appear to be unemployment and inflation. As reported in the table, both correlate a –0.35 with government vote support. However, in a series of multivariate tests, the unemployment variable, as well as the other economic variables, manifested little if any independent effect. The survivor variable was inflation, which carries an even larger coefficient once supressor effects are removed via multivariate analysis.[6]

Table 1
Correlations between the British economy and the vote[a]

	Vote	Unemployment Rate	Inflation Rate	Interest Rate	Growth	Misery Index
Vote	1.00					
Unemployment	–0.35	1.00				
Inflation	–0.35	–0.22	1.00			
Interest Rate	–0.57	0.42	0.67	1.00		
Growth	0.09	–0.27	–0.43	–0.33	1.00	
Misery Index	–0.56	0.59	0.66	0.87	–0.57	1.00

[a] Note: $N = 12$; all economic variables are lagged six months (see Appendix).

[5] For a review of this literature, see Lewis-Beck and Stegmaier (2000: 206).

[6] Interestingly, the case for the forecasting potential of the inflation rate variable appeared in an earlier investigation by Lewis-Beck (1988: 9), predicting these general elections from 1959–1983.

5. Specifying the vote function

Economics, important as it is, is not everything. A standard vote function specification for mature parliamentary systems reads, conceptually, as follows:

$$\text{Government Support} = f(\text{economic, noneconomic issues}) \quad (1)$$

In Europe, the model has received extensive testing in France, most recently to forecast the surprise defeat of the Gaullists in the 1997 National Assembly elections (Fauvelle-Aymar and Lewis-Beck, 1997). The independent variables were GDP growth and presidential approval in the polls, both measured six months before the election. Several things are worth noting. First, the model has one macroeconomic indicator and one executive popularity indicator. That basic political economy specification seems to hold across several democratic polities.[7] Second, the economic and the political variables, as measured, vary somewhat and naturally enough, according to national history. Third, the independent variables each have a six-month lead time. Obviously, this allows for forecasting at a nontrivial distance from the election itself. But that six-month lead specification does not rest on mere opportunistic advantage. In the US, in France, and as well in the British case, predictive power appears maximized at this lead. Specifically, for the UK, the six-month lead yields less prediction error than either nine-month or three-month leads. Reasons for this, perhaps counterintuitive, finding come from the more profound perspective distance affords the voter, coupled with the forthcoming equilibration of the campaign itself.[8]

In Britain, we should expect voters strongly to sanction or reward government, because the line of policy responsibility is so clear. Lewis-Beck and Paldam (2000: 120) observe that, in a two-party dominated parliamentary system, "economic responsibility in a national election is directly assigned." Further, Nadeau et al. (2001) show that economic voting is strongest in Britain, as compared to seven other European countries, because of the clarity of government responsibility as economic manager. The UK, then, affords a textbook test. Bad economics should lead to punishing votes.

But the sanction is not likely to exhaust itself in the economic realm. The government of course must handle many other issues. How to measure its performance on these other issues? The usual approach is an executive approval measure. For example, in the United States, the presidential approval variable "certainly taps noneconomic issues" and can be separated from economic issues with proper statistical controls (Lewis-Beck and Rice, 1992: 46). We apply the same logic to Britain, employing a neglected public opinion measure, which asks respondents whether they approve or disapprove of the "government record." Interestingly, this is conceptually quite close to the American presidential job approval item for popularity, appropriately adapted to the UK parliamentary context. On balance, then, if voters do not

[7] Besides the already cited work on the US and French cases, see also Nadeau and Blais (1993) on Canada and Jérôme et al., (1998) on Germany.

[8] The best discussion of these reasons appears in Campbell (2000).

like how the government is handling the issues, they register disapproval in the survey, going on to vote against in the next election. Across the election series, this government approval variable does correlate at 0.72 with government party vote share. Of course, part of this relationship is due to economic conditions that must be partialed out through multivariate analysis, to which we now turn.

The specification is as follows,

$$V = \mathrm{a} + \mathrm{b}E + \mathrm{c}P + \mathrm{d}T + \mathrm{e} \tag{2}$$

Where V = vote share (in percent) going to the governing party in the general election; E = inflation rate six months prior to the general election; P = public approval (in percent) of the government record; T = number of terms the governing party has been in office; a, b, c, d = parameters to be estimated; e = error term.

Note that, in addition to the core political economy variables, T (number of terms in office) has been included. The notion that “ruling costs votes” has been explored for some Western European countries (Paldam, 1986). Also, it appears as a factor in a leading US presidential election forecasting model, referred to as a “time for a change” variable (Abramowitz, 2000). Because the link between policy responsibility and the voter is rather tight in Britain, it seemed valuable immediately to control for “the cost of ruling”, in order to avoid estimation bias and possibly false conclusions. Below, then, are the ordinary least squares (OLS) estimates for the model as specified:

$$\begin{array}{l} V = -\underset{(8.48)}{42.7} - \underset{(4.74)}{*0.97E} + \underset{(3.30)}{*0.27P} - \underset{(3.18)}{*3.1T} + \mathrm{e} \end{array} \tag{3}$$

R-sq. = 0.88 adj. R-sq. = 0.83 SEE = 2.27

Durbin-Watson = 2.26 N = 12

Where the figures in parentheses are absolute t-ratios; $*$ = statistically significant at 0.01, one-tail ($t > 2.31$); R-sq. = the coefficient of multiple determination; adj. R-sq. = the coefficient of multiple determination adjusted for degrees of freedom; SEE = the standard error of estimate; Durbin-Watson = statistical test for first-order autocorrelation; N = the number of observations, on general elections 1955–1997.

6. Coefficient interpretation and regression diagnostics

The model provides a good fit to the data, accounting for over 80% of the variation in government support. Moreover, the slope coefficients are in the expected direction, and easily statistically significant. The coefficients appear plausible and suggest powerful effects. The economy-election connection shows unit elasticity, with a one percentage point increase in inflation generating a one percentage point decrease in government support. With regard to the approval variable, a four point rise in approval should give the government about a one point increase in vote share. Interestingly, these coefficients, b = -0.97 and c = 0.27, are quite close in raw magnitude to the economy and approval coefficients for a standard retrospective model forecasting the US presidential vote (see Lewis-Beck and Tien, 2000: 90). With respect to relative importance, the inflation variable (standardized coefficient = −0.71) edges out the approval variable (standardized coefficient = 0.50), although both obviously have important effects. Also, the term variable exercises a sizeable impact. For each additional term, the government can expect to lose over 3% of the vote. Obviously, in some sense its continued service is its own undoing, as over time discontented coalitions mobilize against it.

Various diagnostics speak to the robustness of the model. The Durbin-Watson statistic indicates an absence of first-order autocorrelation. Cross-validation tests suggest the model is stable (Beck, 2000). In particular, we reran the model 12 times, omitting each election one-at-a-time. The signs of the three independent variable coefficients never changed, and these coefficients were always statistically significant. As well, the R-squared stayed high, varying from 0.83 to 0.92.[9] Beyond this, other tests gave encouraging results. No studentized residual reached the value “2”. Further, when the regular residuals were plotted against the dependent variable, they were basically evenly split half above zero and half below, revealing no aberrant pattern. Finally, a specification test was run, on the argument that inflation might affect Conservative and Labour support differentially (see Sanders, 2000: 289). Inclusion of this interaction term (Inflation x Incumbent Party) fell far short of conventional significance levels. In sum, the model resists multiple statistical challenges, standing secure in its essentials.

7. Forecasting characteristics of the model

How well does the model forecast? Various assessments are possible. Within the sample itself, the average absolute prediction error is only 1.4 percentage points, which is quite low. However, a tougher test is how it does out-of-sample. In the above cross-validation tests, when each of the elections was omitted in turn, the average absolute forecast error was 2.3 points. That fairly low number would be lower if the unusual second election in 1974 were excluded, with its error of 6.1. Table 2 shows the predicted government party gain (or loss) from these out-of-sample forecasts, compared to the observed government party gain or loss (compare columns 1 and 2). We see that the model accurately foresaw the government’s fortune, in ten of the twelve elections (the predicted signs of the change were wrong only in the October 1974 and the 1979 contests). Put another way, the model can generally tell the government what direction its support is likely to take.

Can we actually forecast which party will take power? Table 3 offers various relevant calculations. The first three columns give the observed government share, the predicted government share and the observed opposition share. The fourth column (1–3) reports the actual government margin against the opposition. The fifth

[9] The adjusted R-squared varied from 0.75 to 0.88.

Table 2
Out-of-Sample Forecasts, 1955–1997

Election	Incumbent party vote Actual	Predicted	Error
1955	49.7	49.8	0.1
1959	49.4	48.6	–0.8
1964	43.4	44.1	0.7
1966	48.0	48.1	0.1
1970	43.1	39.7	–3.4
1974 F	37.9	38.8	0.9
1974 O	39.3	33.2	–6.1
1979	36.9	41.4	4.5
1983	42.4	45.0	2.6
1987	42.3	42.0	–0.3
1992	41.9	37.9	–4.0
1997	30.7	35.2	4.5
Mean absolute error			2.3

Table 3
Actual and predicted election outcomes[a.]

Election	Incumbent	(1) Government observed	(2) Government predicted	(3) Opposition observed	(1–3) Actual margin		(2–3) Predicted margin	
1955	C	49.7	49.8	46.4	3.3	(C)	3.4	(C)
1959	C	49.4	48.6	43.8	5.6	(C)	4.8	(C)
1964	C	43.4	44.1	44.1	–0.7	(L)	0.0	(C/L)
1966	L	48.0	48.1	41.9	6.1	(L)	6.2	(L)
1970	L	43.1	39.7	46.4	–3.3	(C)	–6.7	(C)
1974 F	C	37.9	38.8	37.2	0.7	(L)	1.6	(C)
1974 O	L	39.3	33.2	35.8	3.5	(L)	–2.6	(C)
1979	L	36.9	41.4	43.9	–7.0	(C)	–2.5	(C)
1983	C	42.4	45.0	27.6	14.8	(C)	17.4	(C)
1987	C	42.3	42.0	30.8	11.5	(C)	11.2	(C)
1992	C	41.9	37.9	34.4	7.5	(C)	3.5	(C)
1997	C	30.7	35.2	43.2	–12.5	(L)	–8.0	(L)
Absolute average					6.4		5.7	

[a.] C = Conservative Party, L = Labour Party; letter in parentheses indicates actual or predicted winner. Except for the February 1974 election (see text), a positive margin means the incumbent is (or will be) reelected and a negative margin means the incumbent is (or will be) defeated.

column (2–3) gives the predicted government margin against the opposition. For example, in 1955 the model predicted 49.8% for the government, while the opposition received 46.4%, for a predicted government margin of 3.4 points, an estimate quite close to the actual government margin of 3.3. Generally in the table, the magnitudes of the actual and predicted margins are very similar. Moreover, the predicted margin has the correct sign ten out of twelve times. The two errant cases are 1964, when the model predicted a tie, and October 1974 when a negative prediction met a positive result. Finally, note that while correct prediction of the margin does not automatically predict the party winner, it failed to do so only once, in the very tight February 1974 election (i.e., the Tory lead over Labour was correctly predicted, but Labour came to power).

Recalling the earlier Sanders critique, the supremely difficult thing is to forecast the "unknown future." Specifically, the difficulty is to forecast elections that, unlike those of Table 3, have yet to take place. We have two bits of evidence here. First, the standard error of estimate of the model, SEE = 2.27, indicates that, at least for the typical future election, the model will probably be off a little over two points. That is not bad. It compares favorably to the margin of error from the final preelection vote intention polls. A second bit of evidence we offer is the forecast of the very next general election, assumed to take place in April 2001. To do so, we simply plug in the six-month ahead values on inflation (E), government record (P), and term in office (T), and do the arithmetic.

$$\begin{aligned} V\mathrm{A} &= 42.7 - *0.97\mathrm{E} + *0.27\mathrm{P} - *3.1\mathrm{T} + \mathrm{e} \\ &= 42.7 - 0.97(3.1) + 0.27(37) - 3.1(1) \\ &= 46.5\% \text{ for Labour} \end{aligned} \tag{4}$$

If this forecast holds, then a Labour victory, with an increase of 3.3 points (up from 43.2 in 1997) should take place, assuming they get the seats.

How would this vote share translate into seats? To answer this question, we employ Tufte's (1974: 68) idea of a vote-seat swing ratio, which seems theoretically and empirically superior to the traditional "cubic rule." Following Norris and Crewe (1994), as an aid to statistical control we confine our swing ratio equation to the post-1970 elections, for this OLS result:

$$\begin{aligned} &S = -59.8 + 2.78V + \mathrm{e} \\ &\mathrm{N} = 7 \text{ (elections, } 1974 - 1997) \quad \mathrm{R} - \text{squared} = 0.96 \end{aligned} \tag{5}$$

Then, to forecast the 2001 seats, on the basis of the above vote share estimate yields,

$$S(2001) = -59.8 + 2.78V = -59.8 + 2.78(46.5) = 69\% \text{ for Labour.} \tag{6}$$

This compares favorably to 1997, when Labour got 63% of the seats.

8. Summary and conclusions

It seems possible to formulate, for the British case, a useful forecasting model of the general election result. The model, much in line with accepted political economy work in other advanced democracies, offers a plausible and parsimonious explanation of government vote support. The equation itself fits the data snugly, has substantial lead time, and promises accurate prediction of the outcome. Of course, no model is guaranteed to work every time. In the US, the best presidential forecasting models fell to the Gore defeat of 2000. This recalls the Sanders (2000) caution, that “unpredictable events”, perhaps an extraordinarily bad campaign, can overturn an apparently invincible forecast.

Appendix A. Data and sources

Election	Vote	Seats	Terms	Approval	Inflation	Unemployment	Misery	Interest	Growth
1955	49.7	54.8	1	53	4.2	1.4	5.6	2.047	4.0
1959	49.4	57.9	2	46	0.0	2.4	2.4	3.253	6.0
1964	43.4	48.3	3	46	2.0	1.8	3.8	4.301	5.3
1966	48.0	57.8	1	49	4.8	1.4	6.2	5.424	2.0
1970	43.1	45.7	2	31	4.7	2.5	7.2	7.700	1.8
1974 F	37.9	46.8	1	29	9.3	1.8	11.1	10.974	7.2
1974 O	39.3	50.2	1	48	15.2	2.0	17.2	11.515	–3.0
1979	36.9	42.4	2	44	8.1	4.2	12.3	11.515	5.1
1983	42.4	61.1	1	37	5.4	10.1	15.5	9.897	–0.1
1987	42.3	57.8	2	34	3.7	11.0	14.7	10.659	3.9
1992	41.9	51.6	3	33	3.7	8.8	12.5	9.860	–0.2
1997	30.7	25.0	4	18	2.7	7.9	10.6	6.020	2.8

Note: All independent variables are lagged six months.

Sources

Vote, Seats & Terms

Electoral data taken from Mackie and Rose (1991, 1997).

Approval of record

Gallup Polls. Question wording: “Do you approve or disapprove of the Government’s record to date?” (% approving). Data for the period 1954–1994 taken from Butler and Butler (1994: 249–259); data for the period 1994–2001 taken from various issues of the Gallup Political and Economic Index.

Inflation

Office for National Statistics. Series used is CZBH: annual change in retail price index (% monthly). Available on the web from StatBase at http://www.statistics.gov.uk

Unemployment

Office for National Statistics. Series used is MGSX: monthly unemployment rate (% seasonally adjusted). Available on the web from StatBase at http://www.statistics.gov.uk Complementary data taken from various issues of the *Employment Gazette*.

Misery Index

Computed by adding the inflation rate and the unemployment rate.

Interest Rate

Bank of England. Series used is KDMM: three-month Treasury Bill rate (monthly). Data taken from various issues of the *Annual Abstract of Statistics*.

Growth

Office for National Statistics. Series used is: annual change in real personal disposable income (%). Data taken from various issues of *Economic Trends*.

References

Abramowitz, A.I., 2000. Bill and Al's excellent adventure: forecasting the 1996 presidential election. In: Campbell, J.E., Garand, J.C. (Eds.), Before the Vote: Forecasting American National Elections. Sage, Thousand Oaks, CA, pp. 47–56.

Beck, N., 2000. Evaluating forecasts and forecasting models of the 1996 presidential election. In: Campbell, J.E., Garand, J.C. (Eds.), Before the Vote: Forecasting American National Elections. Sage, Thousand Oaks, CA, pp. 161–167.

Butler, D., Butler, G., 1994. British Political Facts 1900-1994. , seventh ed. St-Martin's Press, New York.

Campbell, J.E., 2000. The science of forecasting presidential elections. In: Campbell, J.E., Garand, J.C. (Eds.), Before the Vote: Forecasting American National Elections. Sage, Thousand Oaks, CA, pp. 169–187.

Campbell, J.E., Garand, J.C., 2000. Before the Vote: Forecasting American National Elections. Sage, Thousand Oaks, CA.

Clarke, H.D., Stewart, M.C., 1994. Economic evaluations and election outcomes: an analysis of alternative forecasting models. In: Broughton, D. (Ed.), British Elections and Parties Yearbook 1994. Frank Cass, London, pp. 73–91.

Fauvelle-Aymar, C., Lewis-Beck, M.S., 1997. L'Iowa donne l'opposition gagnante, Libération, May 23 (#4978), p.15.

Jérôme, B., Lewis-Beck, M.S., Jérôme-Speziari, V., 1998. Allemagne: prévisions politico-économiques et élections législatives de septembre 98, Le Figaro, July 10.

Lewis-Beck, M.S., 1988. Economics and Elections: The Major Western Democracies. University of Michigan Press, Ann Arbor, MI.

Lewis-Beck, M.S., 2001. Modèles économétriques du vote. In: Perrineau, P., Reyne, D. (Eds.), Dictionnaire du vote. Presses Universitaires de France, Paris.

Lewis-Beck, M.S., Rice, T.W., 1992. Forecasting Elections. Congressional Quarterly Press, Washington, DC.

Lewis-Beck, M.S., Paldam, M., 2000. Economic voting: an introduction. Electoral Studies 19, 113–121.

Lewis-Beck, M.S., Stegmaier, M., 2000. Economic determinants of electoral outcomes. Annual Review of Political Science 3, 183–219.

Lewis-Beck, M.S., Tien, C., 2000. The future in forecasting: prospective presidential models. In:

Campbell, J.E., Garand, J.C. (Eds.), Before the Vote: Forecasting American National Elections. Sage, Thousand Oaks, pp. 83–102.
Mackie, T.T., Rose, R., 1991. The International Almanac of Electoral History. , third ed. Macmillan, London.
Mackie, T.T., Rose, R., 1997. A decade of election results: updating the international almanac. Center for the Study of Public Policy, University of Strathclyde, Glasgow.
Mughan, A., 1987. General election forecasting in Britain: a comparison of three simple models. Electoral Studies 6, 195–207.
Nadeau, R., Blais, A., 1993. Explaining election outcomes in Canada: economy and politics. Canadian Journal of Political Science 26, 775–790.
Nadeau, R., Niemi, R.G., Yoshinaka, A., 2001. A cross-national analysis of economic voting: taking account of the political context across time and nations. Electoral Studies (forthcoming).
Norris, P., Crewe, I., 1994. Did the British marginals vanish? Proportionality and exaggeration in the British electoral system revisited. Electoral Studies 13, 201–221.
Paldam, M., 1981. A preliminary survey of the theories and findings on vote and popularity functions. European Journal of Political Research 7, 1–26.
Paldam, M., 1986. The distribution of election results and the two explanations of the cost of ruling. European Journal of Political Economy 2, 5–24.
Sanders, D., 1991. Government popularity and the next general election. Political Quarterly 62, 235–261.
Sanders, D., 1993. Forecasting the 1992 general election result: the performance of an economic model. In: Denver, D. (Ed.), British Elections and Parties Yearbook. Harvester Wheatsheaf, London.
Sanders, D., 1995. Forecasting political preferences and election outcomes in the UK: experiences, problems and prospects for the next general election. Electoral Studies 14, 251–272.
Sanders, D., 1996. Economic performance, management competence and the outcome of the next general election. Political Studies 44, 203–231.
Sanders, D., 2000. The real economy and the perceived economy in popularity functions: how much do voters need to know? A Study of British Data, 1974-97. Electoral Studies 19, 275–294Tufte, E.R., 1974. Data Analysis for Politics and Policy, Prentice-Hall, Englewood Cliffs, NJ.
Whiteley, P., 1979. Electoral forecasting from poll data: the British case. British Journal of Political Science 9, 219–236.